K Prakash

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.04
T.S.W.R.S COE(Boys) Bellampally	Class XII (Telangana State Board of Intermediate Education)	2022	95.70%
T S S W RESL BOYS CHILKOOR	Class X (Board of Secondary Education Telangana)	2020	9.80%

WORK EXPERIENCE

Full Stack Developer Intern | Rengy

- Worked on backend services and REST APIs using **Node.js, Express, Fastify, PostgreSQL, and MongoDB**.
- Developed automated **WhatsApp notifications** and in-app alerts for project status changes and workflow reminders.
- Implemented dynamic **DOCX/PDF generation** and integrated the **OpenAI API** to extract data from uploaded utility bills.
- Built **AWS S3** document upload/download handling with file validation, preview, and image rotation controls.
- Created advanced **search, filtering, sorting, and pagination** across frontend tables and backend query helpers.
- Refactored database queries and **MongoDB aggregation pipelines** to eliminate duplicate lookups and improve dashboard loading speed.
- Collaborated with frontend and QA teams to fix bugs and deliver UI features using **React, TypeScript, and TanStack Query**.

PROJECTS

Team Management & RBAC System | *Node.js · MongoDB · Redis Pub/Sub · Socket.IO* | [Github Link]

- Built a multi-tenant team management platform with task boards, **real-time messaging**, and **audit logging**, controlled by a two-tier **RBAC engine** that combines Redis caching with dynamic resource grants across 39 permission keys.
- Delivered an end-to-end **workspace administration** system with email-based invitations, role lifecycle management, global/team broadcasts, OpenAPI docs, and automated **integration tests** covering security, validation, JIT access, real-time sync, and task workflows.
- Implemented an atomic **Just-In-Time (JIT) access elevation** workflow with **anti-self-approval** and CAS concurrency controls to eliminate race conditions during simultaneous administrative approvals.
- Optimized system reliability with **Redis fixed-window rate limiting** (30 req/15 min auth, 600 req/min API), **51 compound MongoDB indexes**, and a fault-tolerant Redis Pub/Sub socket adapter.

BleachVerse--Hybrid Search Engine | *Node.js · PostgreSQL · Neo4j · Redis · Docker* | [Github Link]

- Engineered a hybrid search engine combining semantic retrieval, structured filtering, and graph traversal through a **deterministic natural-language query router**.
- Integrated local **all-MiniLM-L6-v2** embeddings with 384-dimensional vectors to index **644 domain documents**, eliminating third-party API costs and rate limits.
- Built database-backed entity resolution with aliases, fuzzy matching, ambiguity safeguards, and strict multi-source intersection across hybrid search queries.
- Modeled complex domain relationships into **Neo4j graph topologies** with automated ETL synchronization from **PostgreSQL** to power multi-hop Cypher traversals.

Redis Server & CLI Client | *C++17. POSIX Sockets. TCP/IP.* | [Github Link]

- Built a multithreaded **in-memory key-value database** and interactive **REPL client** in C++17 with zero external dependencies, handling **concurrent clients** over **TCP/IP** via **POSIX sockets** with **mutex-based thread safety**.
- Implemented a custom **RESP 2 protocol** parser/serializer (Strings, Integers, Arrays, Error frames) alongside core commands (SET, GET, DEL, EXISTS, INCR, DECR, PING, ECHO) backed by **hash-table lookups**.

SKILLS

Programming Languages: C/C++, Python, R, JavaScript, TypeScript

Web Development: Node.js, Express.js, REST APIs, React.js, HTML5, CSS3, Redux, Zustand

Databases: PostgreSQL, MongoDB, Neo4j, Redis

DevOps & Tools: Git, GitHub, Docker, Postman, Linux, AWS

CS Fundamentals: DBMS, Operating Systems, Computer Networks

POSITIONS OF RESPONSIBILITY

Internship Coordinator (OCS)-- Coordinated internship opportunities by managing company outreach, student communication, internship applications, and the execution of internship-related activities.

Media & PR Manager (EML)--Developed and maintained event websites while managing digital platforms, online content, and technical implementation for organizational initiatives.

RELEVANT COURSES

Relevant Coursework: Data Structures & Algorithms, Introduction to Programming, Sequence Alignment Algorithms, Big Data Biology & Biological Databases, Introduction to Artificial Intelligence, Probability & Random Variables, Elementary Linear Algebra, Biostatistics, Calculus I & II

EXTRACURRICULAR

Runner-up--Novartis Discoverathon | *PyTorch, ChemBERTa, RDKit, Scikit-learn, XGBoost*

- Developed an ML pipeline for hERG blockade prediction using **16,650 compounds**, achieving **0.832 ROC-AUC** on **3,331 unseen compounds** with **ensemble models** and **ChemBERTa**.